MOI GIRLS' ELDORET 2025 PREDICTION

POSSIBLE KCSE EXAM

MATHEMATICS

121/2

PAPER 2

TIME: 2½ HOURS

NAME……………………………………………………………………………………

SCHOOL………………………………………………… **SIGN**………………………

INDEX NO……………………………………………… **ADM NO**……………………

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO THE CANDIDATES

a) *Write your **name** and **Random no.** the spaces provided above.*
b) *Sign and write **date** of examination in the spaces provided above.*
c) *This paper consists of **two** sections; **SectionI** and **Section II.***
d) *Answer **All** questions in **Section I** andonly **Five**questions from **section II***
e) *Show all the steps in your calculations giving answers at each stage in the spaces provided below each question.*
f) *Marks may be given for correct working even if the answer is wrong.*
g) *Non-programmable silent electronic calculators and KNEC Mathematical tables may be used except where stated otherwise.*

For examiner's use only.

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	**Total**

Section II

17	18	19	20	21	22	23	24	**Total**

GRAND TOTAL

SECTION I (50 marks)

*Answer **all** the questions in this section in the spaces provided.*

1. Evaluate without using tables or calculators. **(3 mrks)**

$$\frac{\log \frac{1}{2} + \log 64}{\log\left(\frac{1}{32} \div \frac{1}{8}\right)}$$

2. Make x the subject of the equation **(3 marks)**

$$\frac{t}{s} = \frac{b}{\sqrt{x-4}}$$

3. Two pipes, P and Q can fill an empty tank in 3 hours and 4 hours respectively. It takes 5 hours to fill the tank when an outlet pipe R is opened the same time with the inlet pipes. Calculate the time pipe R takes to empty the tank. **(3 marks)**

4. Given that $M = i - 3j + 4k$, $W = 6i + 3j - 5k$ and Q = 2M + 5N, find the magnitude of Q to 3 significant figures. **(3 marks)**

5. A triangle ABC is such that a =14.30 cm, b =16.50 cm and B =56º. Find the radius of a circle that circumscribes the triangle. **(3 marks)**

6. Construct a circle centre O and radius 3 cm. Construct two tangents from a point T, 6.5 cm from O to touch the circle at W and X. measure Angle WTX. **(3 marks)**

7. Grace deposited Ksh 16 000 in a bank that paid simple interest at the rate of 14% per annum. Joyce deposited the same amount of money as Grace in another bank that paid compound interest semi- annually. After 4 years, they had equal amounts of money in the banks.

Determine the compound interest rate per annum, to I decimal place, for Joyce's deposit**.(4 marks)**

8. Simplify $\frac{\sqrt{7}}{\sqrt{7-3}}$, leaving your answer in the form $a+b\sqrt{c}$, where a, b, and c are integers.

(2 marks)

9. Solve the equation **(4marks)**

$x-y=1$

$x^2+2y^2=1$

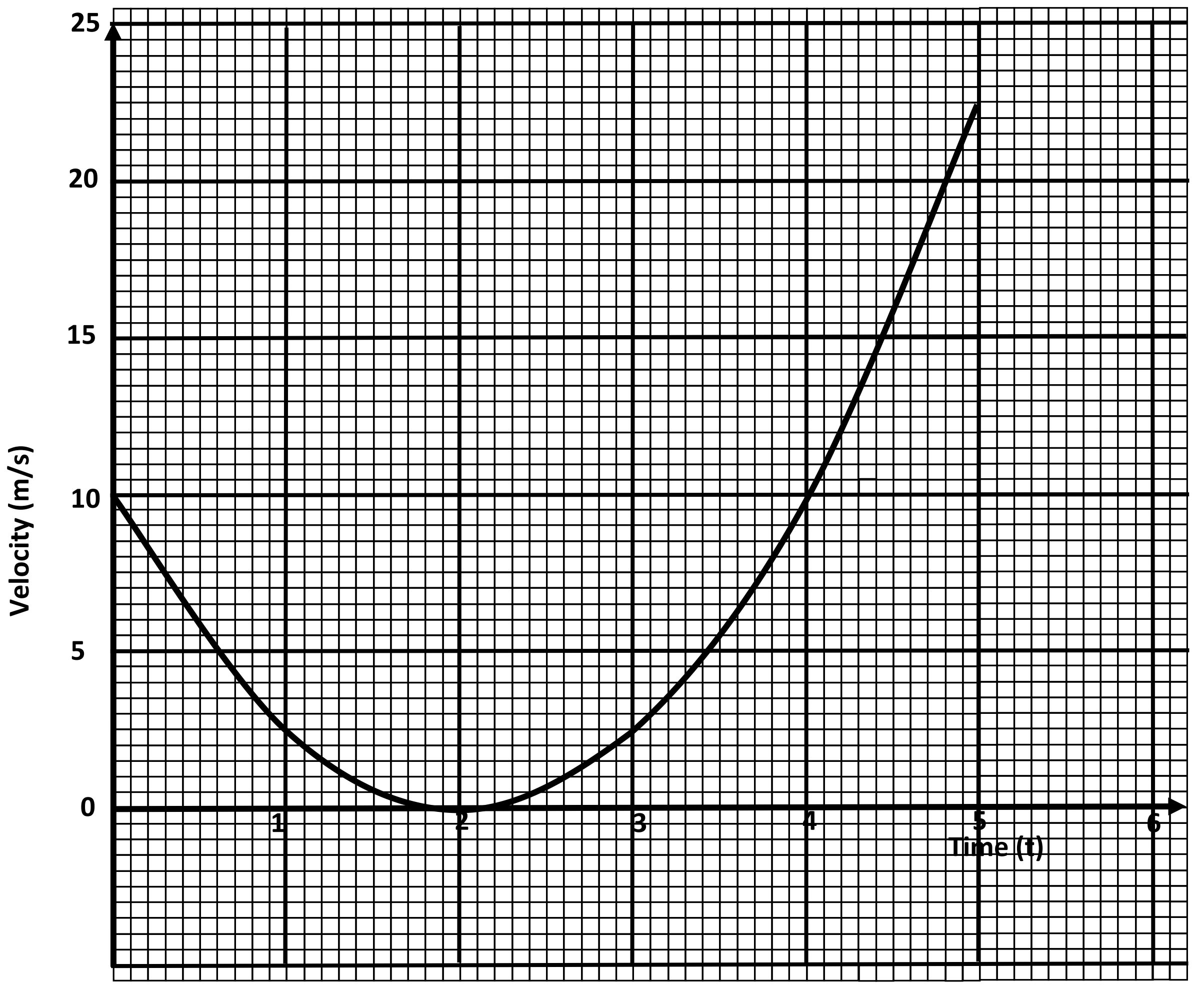
10. Grade I coffee cost sh 500 per kilogram while grade II coffee costs sh 400 per kilogram. The grades are mixed to obtain a mixture that costs sh 420 per kilogram. In what ratio should the two grades be mixed? **(3 marks)**

11. The base length and height of parallelogram were measured as 8.4 cm and 4.5 cm respectively. Calculate the maximum absolute error in the area of the parallelogram. **(3 marks)**

12. (a) Expand $(1 + \frac{1}{2}x)^{10}$up to the fourth term.

(b) Hence, find the value of $(0.84)^{10.}$ **(3 marks)**

13. The graph below shoes the relationship between velocity of a body and time (t) seconds in the interval $0 \leq t = 5$.

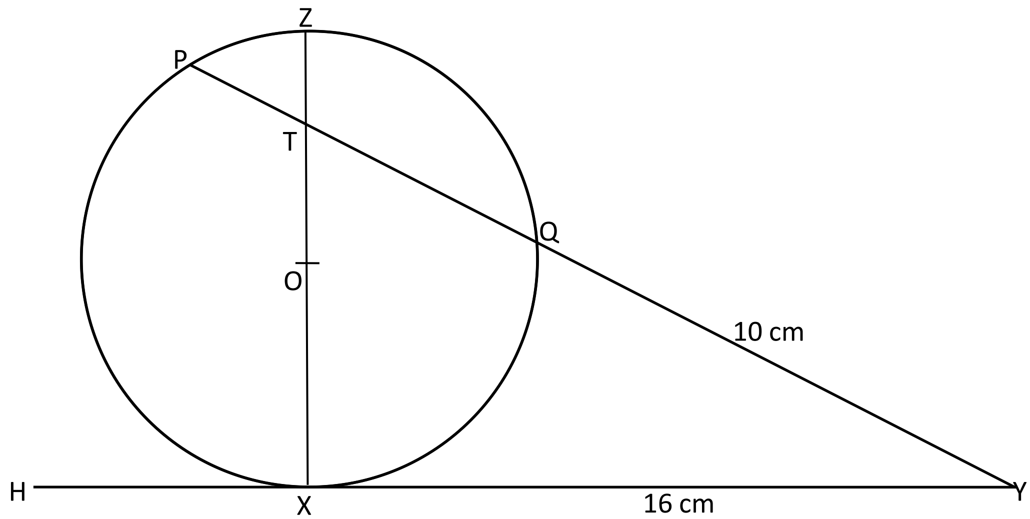


Use the graph to determine ;

(a) the average rate of change of velocity between $t = 2.5$ seconds and $t = 5$ seconds. **(2 mrks)**

(b) the instantaneous rate of change at $t = 4$ seconds. **(2 mrks)**

14. In the figure below, the tangent HXY meets chord PQ produced at Y. Chord XZ passes through the centre, O, of the circle and intersects PQ at T. Line XY = 16 cm and QY = 10 cm.



(a) Calculate the length PQ. **(2 marks)**

(b) If ZT = 4 cm and PT: TQ = 3:5, find XT. **(2 marks)**

15. Quantity P varies partly as Q and partly varies inversely as square of Q. When Q = 1, P = 1 and when Q = ½, P = -3. Find the equation of the relationship connecting P and Q. **(3 marks)**

16. $OA = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} and\ OB = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$. A point Q divides line AB externally the ratio 5:2. Find the position vector of point Q. **(3 marks)**

SECTION II (50 Marks)

Answer any five questions from this section.

17. Two tanks of equal volume are connected in such a way that one tank can be filled by pipe A in 1hour 20minutes. Pipe B can drain one tank in 3hours 36minutes but pipe C alone can drain both tanks in 9 hours. **Calculate**:

(a) The fraction of one tank that can be filled by pipe A in one hour**.**
(2mks)

(b) The fraction of one tank that can be drained by both pipes B and C in one hour.
(4mks)

(c) Pipe A closes automatically once both tanks are filled. Assuming that initially both tanks are empty and all pipes opened at once, calculate how long it takes before pipe A closes. **(4mks)**

18. An examination involves a written test and a practical test. The probability that a candidate passes the written test is $^{6}/_{11}$ if the candidate passes the written test, then the probability of passing the practical test is $^{3}/_{5}$, otherwise it would be $^{2}/_{7}$

(a) **Illustrate** this information on a tree diagram**.**

(2mks)

(b) **Determine** the probability that a candidate is awarded

(i) Credit for passing both tests**.**

(2mks)

(ii) Pass for passing the written test.

(2mks)

(iii) Retake for passing one test.

(2mks)

(iv) Fail for not passing the written test.

(2mks)

19. (a) Conctruct triangle PQR with PQ = 7.2cm, QR = 6.5cm and angle PQR = 48^0 **(3mks)**

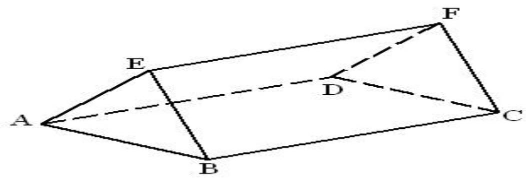
(b) The locus L1, of points equidistant from P and Q, and locus , L2 of points equidistant from P and R, meet at M. Locate M and measure **QM**

(4mks)

(c) A point x moves within triangle PQR such that QX $\geq$ QM. Shade and label the locus of X.

(3mks)

20. The figure below represents a prism with a cross section of an equilateral triangle of side 8cm and length 12cm, as shown below.



(a) Draw the net of the prism ABCDEF **(2mks)**

(b) **Calculate** the angle between the plane ABCD and the line BF.
(2mks)

(c) M is the midpoint of EF. **Calculate**

(i) The length BM
(2mks)

(ii) The perimeter of triangle BMD.
(2mks)

(d) **Calculate** the angle between the plane ABM and the base plane ABCD.

(2mks)

21. Give the matrix $A = \begin{pmatrix} -1 & -4 \\ 1 & 3 \end{pmatrix}$

(a) (i) **Calculate** A^2 and A^3

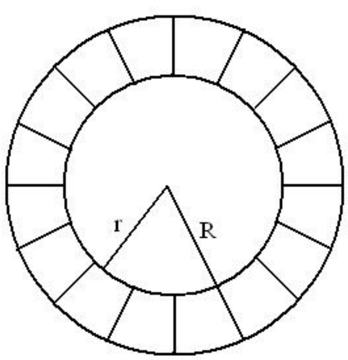
(2mks)

(ii) **Find** the values of the constants p and q for which $A^2 = pA + qI$ where I is the identity matrix**.**

(3mks)

(iii) The triangle ABC maps onto $A^1B^1C^1$ under the transformation represented by matrix A. **Find** the area of triangle ABC if the area of triangle $A^1B^1C^1$ is **21cm²** **(3mks)**

(b) The figure shows two concetric circles such that the ratio of their radii is 1: 3. If the area of the shaded region is 78.4 square units, **calculate** the area of the larger circle. **(2mks)**



22) . A certain uniform supplier is required to supply two types of shirts: one for girls labelled G and the other for boys labelled B. The total number of shirts must not be more than 400. e as to supply more of type G than of type B. However the number of type G shirts must not be more than 300 and the number of type B shirts must not be less than 80. by taking x to be the number of type G shirts and y the number of type B shirts,

(a) **Write down** in terms of x and y all the inequalities representing the information above. **(3mks)**

(b) On the grid provided **draw** the inequalities and shade the unwanted regions. **(4mks)**

(c) Given that type G costs Shs. 500 per shirt and type B costs Shs. 300 per shirt.

(i) Use the graph in (b) above to **determine** the number of shirts of each type that should be made to maximize profit**.**

(1mk)

(ii) **Calculate** the maximum possible profit**.**

(2mks)

23. **(a)** The equation of a curve is given by $y = X^3 + X^2 - bx$. **Show** that the value of X at the minimum turning point is $\frac{-1+\sqrt{19}}{3}$

(3mks)

(b) The displacement X metres of a particle moving along a straight line after t seconds is given by $X = 4t + 2t^2 - t^3$

(i) **Find** its initial acceleration **(2mks)**

(ii) **Calculate** the time when the particle was momentarily at rest. **(2mks)**

(c) (i) Find the values of X where the curve $y = X^2 (x - 2)$ crosses the x-axis. **(1mk)**

(ii) Hence **find** the area enclosed by the curve $y = X^2 (x - 2)$, the lines $x = 0$, $x = 2\frac{2}{3}$ and the x-axis. **(2mks)**

24. The marks of 50 students in a mathematics test were taken from a form 4 class and recorded in the table below.

Mark (%)	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
Frequency	2	5	7	9	11	8	5	3

(a) On the grid provided, **draw** a cumulative frequency curve of the data. **(3mks)**

Take: 1cm to represent 5 students on the vertical scale and 1cm to represent 10 marks on the horizontal scale.

(b) From your curve in (a) above

(i) Estimate the median mark**.**

(1mk)

(ii) Determine the Interquartile deviation**.**

(2mks)

(iii) Determine the 10^{th} to 90^{th} percentile range**.**

(2mks)

(c) It is given that students who score over 45 marks pass the test. Use graph in (a) above to **estimate** the percentage of students that pass**.**

(2mks)